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Q1. (c) i. Claim: if g & h agree on free variables in ϕ , $V_g(\phi) = V_h(\phi)$

Pf: Assume g & h agree on the free variables in ϕ

We prove on induction on the complexity of ϕ

BC: $\phi = \pi \alpha_1 \dots \alpha_n$

Suppose $V_g(\phi) \neq V_h(\phi)$

WLOG, let $V_g(\phi) = 1$, $V_h(\phi) = 0$

Since the valuation function is bivalent

By assumption,

$$\langle [B_1]_g, [B_2]_g, \dots, [\alpha]_g \rangle \in I(\pi)$$

$$\text{while } \langle [B_1]_h, [B_2]_h, \dots, [\alpha]_h \rangle \notin I(\pi)$$

where B_i is a free variable and α_i is either a constant

or a bound variable - How can α_i be bound

This means that the n -ary relation must differ in at least one member when we
are in the
if α_i is a constant then for any g, h , $[\alpha]_g = [\alpha]_h$ are in the
base case?

if α_i is a bound variable, and g, h agree on all free variables,

but differ on α_i , $V_g(\phi) = V_h(\phi)$ since the truth value

of the bound predicate is determined by all ^{possible} variable assignments

Not just one $\rightarrow$ This is first-order predicate
logic, a predicate cannot be bounded
 $\therefore$ it must be the case that for at least one free variable B_i , $[B_i]_g \neq [B_i]_h$

By contraposition, claim is proved.

IH: Assume that $V_g(\psi) = V_h(\psi)$ for all $C(\psi) < C(\phi)$

We have three cases

1) $\phi = \sim \psi$

Since $V_g(\psi) = V_h(\psi)$ by IH,

$$V_g(\sim \psi) = V_h(\sim \psi)$$

$$V_g(\phi) = V_h(\phi)$$

2) $\phi = \psi_1 \rightarrow \psi_2$

By IH, $V_g(\psi_1) = V_h(\psi_1)$ &

$$V_g(\psi_2) = V_h(\psi_2)$$

So, if $V_g(\psi_1) = 1, V_g(\psi_2) = 0$

$$V_g(\psi_1 \rightarrow \psi_2) = V_h(\psi_1 \rightarrow \psi_2) = 0$$

else,

$$V_g(\psi_1 \rightarrow \psi_2) = V_h(\psi_1 \rightarrow \psi_2) = 1$$

$$\therefore V_g(\phi) = V_h(\phi)$$

3) $\phi = \forall \alpha \psi$

By IH, $V_g(\psi) = V_h(\psi)$

Then $V_g(\forall \alpha \psi) = 1$ iff for every $u \in D$, $V_{g_u}(\psi) = 1$

iff $\forall u \in D$, $V_{h_u}(\psi) = 1$ iff $V_h(\forall \alpha \psi) = 1$

$$\therefore V_g(\forall \alpha \psi) = V_h(\forall \alpha \psi)$$

$$\Rightarrow V_g(\phi) = V_h(\phi) \quad \text{IH proven}$$



ii. Claim: if $[\alpha]_g = [\beta]_g$, then $V_g(\phi) = V_g(\phi(\beta/\alpha))$

Suppose for any model M ,
 Define $\phi = \pi \alpha \delta_1 \dots \delta_n$

You can't
 assume that

$\phi = \pi \alpha \delta_1 \dots \delta_n$,
 it might be a
 complex formula

$$V_g(\phi) = 1 \text{ iff } \langle [\alpha]_g, [\delta_1]_g, \dots, [\delta_n]_g \rangle \in I(\pi)$$

$$\text{iff } \langle [\beta]_g, [\delta_1]_g, \dots, [\delta_n]_g \rangle \in I(\pi) \text{ since } [\alpha]_g = [\beta]_g$$

$$\text{iff } V_g(\phi(\beta/\alpha)) = 1$$

$$\therefore V_g(\phi) = 1 \text{ iff } V_g(\phi(\beta/\alpha)) = 1$$

$$\text{since } V_g: F \rightarrow \{0, 1\}$$

thus reduces to

$$V_g(\phi) = V_g(\phi(\beta/\alpha))$$

(b) i. $\models_{PC} \forall \alpha \phi \rightarrow \phi(\beta/\alpha)$

AFC that $V_{M,g}(\forall \alpha \phi \rightarrow \phi(\beta/\alpha)) = 0$ for some $M = \langle D, I \rangle$ & variable
 assignment g for M

$$\therefore \text{Dropping } M, V_g(\forall \alpha \phi) = 1 \text{ \& } V_g(\phi(\beta/\alpha)) = 0$$

let g be some variable assignment such that

$$[k]_{g \cup u}^{\alpha \beta} = [\alpha]_{g \cup u}^{\alpha \beta}$$

$$\text{Then, if } V_{g \cup u}^{\alpha \beta}(\phi(\beta/\alpha)) = V_{g \cup u}^{\alpha \beta}(\phi) = 0$$

~~X~~ But if $V_g(\forall \alpha \phi) = 1$, then for every $u \in D$, $V_{g_u}^\alpha(\phi) = 1$.
 Yes, but you should show $V_{g_u}^\alpha(\phi) \geq 0$, which does not contradict.

ii. $\models_{pc} \forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)$ whenever α does not occur free in ϕ

AFC that there is an M st. $V_{M,g}(\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)) = 0$ for some g for M

$$\therefore V_g(\forall \alpha (\phi \rightarrow \psi)) = 1, V_g(\phi \rightarrow \forall \alpha \psi) = 0$$

$$\therefore V_g(\phi) = 1, V_g(\forall \alpha \psi) = 0$$

$$\therefore \text{there is some } u \in D \text{ st. } V_{g_u}^\alpha(\psi) = 0$$

$$\therefore \text{for all } u \in D, V_{g_u}^\alpha(\phi \rightarrow \psi) = 1$$

we set $u = u'$

$$\therefore \text{Since } \alpha \text{ does not occur freely in } \phi, V_{g_{u'}}^\alpha(\phi) = 1$$

$$\therefore V_{g_{u'}}^\alpha(\psi) = 1 \quad \times$$

iii. If $\models_{pc} \phi$, then $\models_{pc} \forall \alpha \phi$

Pf. Assume $\models_{pc} \phi$, then for any M , and any variable assignment g for M ,

$$V_{M,g}(\phi) = 1$$

AFC that $\not\models_{pc} \forall \alpha \phi$. So, there is some M st. $V_{M,g}(\forall \alpha \phi) = 0$

$$\therefore \text{there is some } u \in D \text{ st. } V_{g_u}^\alpha(\phi) = 0 \text{ under that } M$$

But for any M , and any g for M , $V_{M,g}(\phi) = 1$

$$\text{so, } V_{g_u}^\alpha(\phi) = 1 \quad \times$$

Q2. (a) $\exists x (Kx \wedge \forall y (Ky \rightarrow y=x) \wedge Bx)$ $\times$

$\exists x (Kx \wedge Bx)$ This says that the present King of France is bald
(I will not repeat this in subsequent questions)

(b) $B_{1x} Kx$

$$V_{M, f}(\exists x (Kx \wedge \forall y (Ky \rightarrow y=x) \wedge Bx)) = 1$$

iff

$$V_{M, g_u^x}(Kx \wedge \forall y (Ky \rightarrow y=x) \wedge Bx) = 1 \text{ for some } u \in D$$

iff

$$V_{M, g_u^x}(Kx) = 1 \text{ and } V_{M, g_u^x}(\forall y (Ky \rightarrow y=x)) = 1$$

$$\text{and } V_{M, g_u^x}(Bx) = 1$$

$$\text{iff } V_{M, g_{uu}^{xy}}(Ky \rightarrow y=x) = 1 \text{ for all } u' \in D, V_{M, g_{uu}^{xy}}(Kx) = 1,$$

$$V_{M, g_{uu}^{xy}}(Bx) = 1$$

$$\text{iff } \{ \langle u \rangle \} = \perp(K), V_{M, g_{uu}^{xy}}(Kx) = 1, V_{M, g_{uu}^{xy}}(Bx) = 1$$

$$\text{iff } \alpha Kx \text{ is defined and } \langle [\alpha Kx]_{g_{uu}^{xy}} \rangle \in \perp(B)$$

$$\text{iff } V_{M, f}(B_{1x} Kx) = 1$$

✓

$$(c) \quad \lambda_x (I_x \wedge B_x) \cup K_y$$

$$(I_x \wedge B_x)^{M_{g,x}} = \{u \in D : V_{M,g_x^1}(I_x \wedge B_x) = 1\}$$

$$V_{M,g}(\lambda_x(I_x \wedge B_x) \cup K_y) = 1 \quad \text{iff}$$

$$\langle [K_y]_{g_y} \rangle \in \lambda_x(I_x \wedge B_x)^{M_{g,x}} \quad \text{iff}$$

$$\text{iff } \{\langle u \rangle\} = I(K) \text{ and } \langle u \rangle \in (I_x \wedge B_x)^{M_{g,x}}$$

$$\text{iff } \{\langle u \rangle\} = I(K) \text{ and } \langle u \rangle \in I(K) \text{ and } \langle u \rangle \in I(B)$$

$$\text{iff } \omega(x) \text{ is defined and } \langle [K]_{g_x} \rangle \in I(B)$$

$$\text{iff } V_{M,g}(B_{\omega(x)}(x)) = 1$$

I read the question as showing equivalence with the other answer from a and this.

(d) The formalisation in (a) does violence to our grammar, by embedding conjunctions & conditions in the statement 'The King of France is bald' to read "there exists an x st. (x is the King of France) and for all y (if y is the King of France then y is identical to x) and (x is bald)"

While the formalisation in (c) is more intuitive to read

"there is a unique y st. it is an x that is the King of France & is bald"

; the imposition of uniqueness via a complex predicate makes the formalisation more natural

But there is no semantic difference — if anything, I prefer L_2 because it is simpler ✓

Q3. (a)

$$V_{M,g}((\exists \alpha : \phi) \psi) = 1 \text{ iff } |\phi^{M,g,\alpha} \cap \psi^{M,g,\alpha}| = 0$$

"there is no ϕ that is ψ "

$$\neg \exists x (\phi \wedge \psi)$$

$$V_{M,g}(\neg \exists x (\phi \wedge \psi)) = 1 \text{ iff}$$

$$V_{M,g}(\exists x (\phi \wedge \psi)) = 0 \text{ iff}$$

$$\text{there is no } u \in D \text{ s.t. } V_{M,g_u^\alpha}(\phi \wedge \psi) = 1$$

$$\text{iff } (\phi \wedge \psi)^{M,g,\alpha} = \emptyset$$

$$\text{iff } |\phi^{M,g,\alpha} \cap \psi^{M,g,\alpha}| = 0$$

✓

$$(b) V_{M,g}((\text{at least } 2 \alpha : \phi) \psi) = 1 \text{ iff } |\phi^{M,g,\alpha} \cap \psi^{M,g,\alpha}| \geq 2$$

"there are at least 2 ϕ that are ψ "

$$\exists x \exists y (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y)$$

$$V_{M,g}(\exists x \exists y (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y)) = 1$$

$$\text{iff there is some } u \in D \text{ st. } V_{M,g}(\exists y (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y)) = 1$$

$$\text{iff there is some } u' \in D' \text{ st. } V_{M,g}(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y) = 1$$

$$\text{where } D' = \{u : u \in D \text{ st. } V_{M,g}(\exists y (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y)) = 1\}$$

$$\text{iff } (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \neg x=y)^{M,g,x} \supseteq \{u, u'\}$$

$$\text{iff } |\phi^{M,g,x} \cap \psi^{M,g,x}| \geq 2 \quad \checkmark$$

$$(c) V_{M,g}((\text{finitely many } \alpha : \phi) \psi) = 1 \text{ iff } |\phi^{M,g,\alpha} \cap \psi^{M,g,\alpha}| \text{ is finite}$$

Let ϕ_n represent "there are at least n things". Now, $\text{At } V_{M,g}(\sim \phi_\infty) = 1 \text{ iff } |\phi_\infty^{M,g,\alpha}| < \aleph_0$

τ is notation

$$\therefore V_{M,g}(\phi_\infty) = 1 \text{ iff } |\phi_\infty^{M,g,\alpha}| \geq \aleph_0 \quad \checkmark$$

Then, suppose $T = \{\phi_1, \dots, \phi_n\}$ and $T \models \phi_\infty$ already indicates that

By compactness, $\exists T_0 \subseteq T$ st. $T_0 \models \phi_\infty$

but T_0 has a largest ϕ_n , so $T_0 \not\models \phi_\infty$

τ is finite.

write $\tau = \{ \phi_n : n \in \mathbb{N} \}$

$$(d) \quad V_{M,g}((\text{most } \alpha: \phi)\psi) = 1 \text{ iff } |\phi^{M,g,\alpha} \cap \psi^{M,g,\alpha}| = |\phi^{M,g,\alpha} - \psi^{M,g,\alpha}|$$

"(most $\alpha: \phi$) ψ " cannot be expressed in $L =$ either since consider some case in

which there are infinite numbers

? Probably Louheim - Skolem

X

$$(b) \quad (\text{No } \alpha: \phi)\psi$$

$\Rightarrow \sim \exists \phi \exists \alpha (\phi \wedge \psi) \Rightarrow$ there is no ϕ such that there exists an α that satisfies both ϕ & ψ
which is the same as there is no ϕ that is ψ

$$(\text{at least two } \alpha: \phi)\psi$$

The same 1-st order answers should work in 2nd order

$\Rightarrow$